

Nurein Umeya
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GitHub: github.com/cl0udsw1tch

PROFESSIONAL SUMMARY

Software engineer and strong with algorithms, mainly those in general computer science, path planning, machine learning, and state estimation.

TECHNICAL SKILLS

Use purely UNIX environments, or anything conforming to the POSIX standard.

- **Programming:** C++, Python, Java, Golang, TypeScript, Bash, SQL
- **Machine Learning & AI:** PyTorch, TensorFlow
- **Cloud & DevOps:** AWS, GitLab CI/CD
- **Robotics & Embedded Systems:** ROS, SLAM, State Estimation, Nvidia CUDA
- **Web & Software Development:** React, Node.js, Express.js, Next.js, Flask, Svelte

EXPERIENCE

Vim-Ticker

- Stock ticker-tape and ticker-chart plugin for the vanilla Vim terminal editor, with live updates
- Implements streams and iterators from scratch to store and process incoming price data
- MVC architecture allowing code readability and future development

Software Engineer Intern | LeddarTech | Sept 2023 – May 2024

- Developed real-time road lane detection for motor vehicles using C++ and computer vision techniques.
- Optimized performance to achieve 15 FPS on embedded Nvidia devices.
- Utilized CI/CD tooling (Bitbucket, Git, Docker) in a Linux development environment.
- Performed extensive code reviews and debugging using GDB.

Teaching Assistant – Mathematics | University of Toronto | Sept 2021 – May 2022

- Led tutorials and graded assignments for Linear Algebra, Multivariable Calculus, and Vector Calculus courses.
- Provided tutoring and mentorship to students outside of lecture hours.

EDUCATION

Master of Engineering (Mechanical – Robotics & ML/Data Science) | University of Toronto | Jan 2023 – July 2025

- Relevant Coursework: Path Planning, Perception, Machine Learning, Big Data Analytics, State Estimation

Bachelor of Science (Neuroscience & Physiology, Mathematics Minor) | University of Toronto | Sept 2017 – Apr 2022

ASSOCIATIONS & EXTRACURRICULARS

- **Neuroscience Association for UofT Students (NAUS) (2020 – 2021)**
 - Organized outreach events, including a talk by Geoffrey Hinton on backpropagation.